Predictive Football Analytics Using Machine Learning

Internal Assessment Project Report

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Abstract

Predicting a football player's performance after a transfer is difficult because the observed outcome reflects both the player's prior characteristics and the environment of the destination club. This project investigates whether destination-team context improves predictions of post-transfer attacking performance beyond a player-only baseline. The target variable, `post_GA90`, represents post-transfer goals plus assists per 90 minutes. Three Random Forest regression configurations are compared: Model A uses player characteristics; Model B adds basic destination-team context; and Model C incorporates richer team and tactical context.

In the principal comparison, all three models are evaluated on the same 1,194 test records. Model A obtains a mean absolute error (MAE) of 0.123159, root mean squared error (RMSE) of 0.172344, and coefficient of determination (R^2) of 0.551043. Model B achieves the lowest MAE (0.121047), RMSE (0.169027), and highest R^2 (0.568156). Model C records MAE 0.121325, RMSE 0.169764, and R^2 0.564382. The reported paired comparison between A and B indicates lower absolute error for Model B (Wilcoxon signed-rank $p = 0.001579$; bootstrap 95% confidence interval for the paired MAE difference: [0.000764, 0.003406]). The comparison between B and C does not show a statistically significant difference in paired absolute error ($p = 0.422961$; 95% confidence interval: [−0.001564, 0.001006]).

The findings suggest that basic destination-team context adds predictive value in the principal experiment, while richer context does not demonstrate a further statistically detectable improvement over Model B. A separate corrected-audit experiment is documented separately because it uses a different test cohort. The results should be interpreted with caution pending full verification of feature timing, the exact final dataset, and the statistical resampling procedure.

Keywords: football analytics, player transfers, Random Forest regression, post-transfer performance, team context, MAE, predictive modelling.

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Chapter 1

Introduction

1.1 Background

Football clubs make recruitment decisions under uncertainty. A player's historical statistics, age, position, playing time, and market valuation provide useful evidence, but they do not fully describe the environment in which the player will perform after moving to another club. The destination team's recent performance, tactical approach, managerial context, and competition for playing time may also be relevant.

Sports analytics has increasingly used structured data and machine-learning methods to describe player actions, evaluate performance, and support recruitment decisions. For example, action valuation work demonstrates that contributions can extend beyond goals and assists alone [2]; machine-learning approaches have also been used in recruitment and player evaluation [3]. Research on tactical analysis highlights the importance of representing team behaviour using measurable features rather than relying only on broad qualitative labels [5, 6, 7].

The present project focuses on a narrower question: whether destination-team context improves prediction of an observable post-transfer outcome when compared with a model that uses player characteristics alone.

1.2 Problem statement

A player may perform differently after transferring because of changes in playing time, role, tactical system, teammates, competition, and adaptation to a new club. A model built only from player-level information may not represent all of these contextual differences. At the same time, adding more features does not guarantee improved generalisation. Contextual variables may be noisy, incomplete, or redundant.

The problem addressed in this project is therefore empirical: compare progressively expanded feature sets under a common evaluation design and determine whether the

additional destination-team information improves test-set predictions.

1.3 Research question

Does incorporating destination-team context improve the prediction of football players' post-transfer performance, measured by `post_GA90`, compared with a model using player characteristics alone?

1.4 Aim and objectives

The aim is to evaluate the incremental predictive value of destination-team context for post-transfer attacking performance.

The objectives are:

1. Construct a transfer-level prediction target based on post-transfer goals and assists per 90 minutes.
2. Establish a player-characteristics-only baseline (Model A).
3. Add basic destination-team context (Model B) and compare it with the baseline.
4. Evaluate a richer contextual configuration (Model C) to test whether it improves on Model B.
5. Compare the models using MAE, RMSE, and R^2 on a common test sample.
6. Examine paired absolute-error differences using bootstrap confidence intervals and Wilcoxon signed-rank tests.

1.5 Research hypotheses

H1 Model B will have lower test-set MAE than Model A.

H2 Model C will have lower test-set MAE than Model B.

H3 The paired prediction-error differences will be assessed statistically rather than inferred from aggregate metrics alone.

H1 and H2 are directional hypotheses. The analysis does not assume that richer feature sets must improve performance.

1.6 Scope and contribution

The contribution is an empirical comparison of player-only, basic-context, and richer-context feature configurations for post-transfer performance prediction. It is not a claim that machine learning, football performance modelling, or tactical analysis is new. The project's focus is whether destination-team context adds measurable predictive value in this particular transfer-performance setting.

Chapter 2

Literature Review

2.1 Machine learning in football performance analysis

Machine learning has been applied to several football analytics tasks, including player evaluation, action valuation, recruitment, and market-value estimation. Decroos et al. developed a framework for valuing player actions beyond the direct outcomes recorded in conventional statistics [2]. Barron et al. studied the use of artificial neural networks in professional soccer recruitment [3]. These studies illustrate how structured performance information can support analytical decision-making, although their objectives differ from predicting an individual player's post-transfer goals and assists per 90 minutes.

Random Forests provide a flexible ensemble method for regression and classification [1]. They can model nonlinear relationships and interactions without requiring a linear functional form to be specified in advance. This makes the method a reasonable baseline for heterogeneous transfer-level features. Nevertheless, a flexible model can overfit, so evaluation on temporally separated data is important.

2.2 Player valuation and transfer-related modelling

Transfer-related studies have used machine learning and performance indicators to estimate player market value or transfer fees. McHale and Holmes examined transfer-fee estimation using performance metrics and machine learning [4]. Such work is relevant because it demonstrates the use of structured player and transfer information in football analytics. However, a financial outcome such as transfer fee is not equivalent to actual performance after a player joins a destination club. The target and evaluation design in this project are therefore distinct from market-value prediction.

2.3 Team tactics and context

Team behaviour can be represented through measurable indicators such as passing, progression, formation, pressing, and attacking or defensive patterns. Fernández-Navarro et al. analysed attacking and defensive styles of play across elite teams [5]. Rein and Memmert discussed the opportunities and challenges associated with big data and tactical analysis in elite football [6]. Goes et al. examined team performance modelling using tactical features derived from tracking data [7]. Together, this body of work motivates representing team context quantitatively.

However, team-style modelling and tactical analysis do not automatically establish that such features improve prediction of a player's realised performance after a transfer. The value of these features depends on their availability, reliability, and relationship to the selected target.

2.4 Player–team fit and the research gap

Recruitment and player-team fit involve more than a player's isolated historical statistics. A player's opportunities and role may depend on the receiving team's structure and tactical environment. Yet scouting suitability, transfer-fee prediction, and post-transfer outcome prediction are different tasks. A model that ranks a player's fit for a team is not necessarily a model that predicts subsequent goals and assists.

The project addresses this distinction by comparing three feature configurations on a shared test sample. The primary question is not whether context is theoretically relevant, but whether the supplied destination-team features improve out-of-sample prediction in the observed experiment.

2.5 Summary

The literature motivates the combination of player performance data and team context, while also showing that model performance must be evaluated against a clear outcome. The present study contributes a focused empirical comparison of progressively richer feature sets. The literature references above should be checked against the original publications and the final institutional citation requirements before formal submission.

Chapter 3

Data and Feature Construction

3.1 Data sources and dataset versions

The project materials describe player and transfer data sourced from Transfermarkt-style tables, including player, transfer, appearance, and valuation records. Destination-team information was also constructed from StatsBomb Open Data in earlier data-preparation work. The supplied research draft records an earlier player-side dataset of 16,862 transfer records and a tactical subset of 3,752 transfer records. These figures describe an earlier construction stage and should not be treated as the verified size of the final modeling dataset.

The project summary reports a later consolidated dataset of 5,969 transfer records, 3,399 unique players, 396 destination clubs, and 74 columns. Its reported date range is 3 September 2012 to 10 June 2025. These descriptive statistics remain subject to recalculation from the final consolidated CSV, which was not included in the available support archive. They are therefore recorded as reported project metadata rather than independently verified facts.

3.2 Prediction unit and target

The prediction unit is a transfer record. The target variable is `post_GA90`, defined as:

$$\text{post_GA90} = \frac{\text{post-transfer goals} + \text{post-transfer assists}}{\text{post-transfer minutes}} \times 90.$$

This represents the player's combined goals and assists per 90 minutes during the specified post-transfer observation window. The earlier data-preparation notes describe a 365-day window after the transfer and a minimum post-transfer playing-time eligibility rule. The final eligibility filters should be confirmed against the final dataset and training script before submission.

The previously reported target summary is: mean 0.257654, median 0.178100,

standard deviation 0.252060, minimum 0, and maximum 1.867220. These statistics require final-dataset verification.

3.3 Feature groups

The three principal models are defined conceptually as follows.

Table 1: Principal feature configurations

Model	Feature group	Purpose
A	Player characteristics and pre-transfer performance	Establish a player-only baseline.
B	Model A features plus basic destination-team context	Test incremental value from basic team information.
C	Model B features plus richer team/tactical context	Test whether richer context adds further predictive value.

The feature names and definitions should be taken from the final training script. In the main model description supplied for this report, player features include age at transfer, height, position, preferred foot, pre-transfer appearances, minutes, goals, assists, per-90 rates, cards, market value, and transfer fee. Contextual features include team performance summaries and, for the richer configuration, additional tactical or team-level information. The precise feature list differs across project iterations; this report does not assume that every candidate feature was included in every model.

3.4 Preprocessing

The supplied main training-script description specifies median imputation for numerical variables and most-frequent imputation for categorical variables, followed by one-hot encoding of categorical predictors. Preprocessing should be fitted on the training data only, with the same fitted transformations applied to validation and test data. This prevents statistics from the held-out test sample from influencing preprocessing.

The report should be read alongside the exact script version used to generate the principal A/B/C predictions. Other supplied scripts implement different experiments and should not be assumed to share identical preprocessing or feature sets.

3.5 Chronological evaluation split

The main training-script description specifies a chronological 60% training, 20% validation, and 20% testing split after sorting records by transfer date. A chronological split is appropriate for examining performance on later transfer records, because it reduces the risk of evaluating on earlier records while training on later records. However, the exact final dataset, split boundaries, and all preprocessing operations have not been independently reconstructed from the original data in this report. The split is therefore described as specified by the supplied script, not as a fully reproduced pipeline.

Chapter 4

Modelling and Evaluation

4.1 Random Forest regression

The main A/B/C experiment is described as using Random Forest regression. The supplied training-script details specify 300 trees, maximum depth 12, minimum leaf size 10, and random seed 42. These settings are reported as the configuration associated with the main comparison; separate tuning or audit scripts may use different settings.

Random Forest regression combines predictions from multiple decision trees. The ensemble can capture nonlinear associations between player characteristics and the target and can accommodate mixed predictor types after preprocessing. Its flexibility also means that training performance may exceed held-out performance, so generalisation must be assessed on validation and test data.

4.2 Evaluation metrics

The study uses three regression metrics.

Mean Absolute Error

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

MAE is the mean absolute difference between observed and predicted target values. Lower values indicate smaller average absolute errors.

Root Mean Squared Error

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

RMSE gives greater weight to larger errors because the residuals are squared before averaging. Lower values indicate better performance under this metric.

Coefficient of determination

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

Under this definition, R^2 compares the model's squared error with the error from predicting the test-set mean. Higher values indicate better fit relative to that baseline. An R^2 value is not percentage prediction accuracy.

4.3 Paired error comparisons

For each test record i and model m, define absolute error as

$$e_{i,m} = |y_i - \hat{y}_{i,m}|.$$

The paired mean absolute-error difference for models A and B is

$$\Delta_{A,B} = \frac{1}{n}\sum_{i=1}^{n}(e_{i,A} - e_{i,B}).$$

A positive difference favours the second model because the first model has the larger absolute error. Pairing is meaningful when models are evaluated on the same records and the observed targets are aligned.

The supplied statistical outputs include bootstrap confidence intervals for paired MAE differences and Wilcoxon signed-rank tests. These procedures provide evidence about whether the observed paired error differences are distinguishable from zero under their respective assumptions. The exact bootstrap resampling unit, number of resamples, confidence-interval method, and any multiple-comparison adjustment still require confirmation from the analysis code. The p-values should therefore be interpreted alongside these methodological caveats.

Chapter 5

Results

5.1 Principal A/B/C comparison

The saved test-prediction files for Models A, B, and C were reported as containing 1,194 aligned records, with matching transfer identifiers and observed target values. Recalculation from these saved predictions reproduced the displayed MAE, RMSE, and R^2 values to the reported precision. This is the principal result set for the report.

Table 2: Principal test-set performance (common test sample)

Model	**Test** N	**MAE** $\downarrow$	**RMSE** $\downarrow$	R^2 $\uparrow$
A: Player only	1,194	0.123159	0.172344	0.551043
B: Player + team	1,194	**0.121047**	**0.169027**	**0.568156**
C: Rich context	1,194	0.121325	0.169764	0.564382

Model B achieved the best result across all three reported test metrics. Compared with Model A, Model B reduced MAE by approximately 1.71% and RMSE by approximately 1.92%. Its R^2 increased from 0.551043 to 0.568156. Model C also improved on Model A, but its MAE and RMSE were slightly higher than those of Model B, and its R^2 was lower.

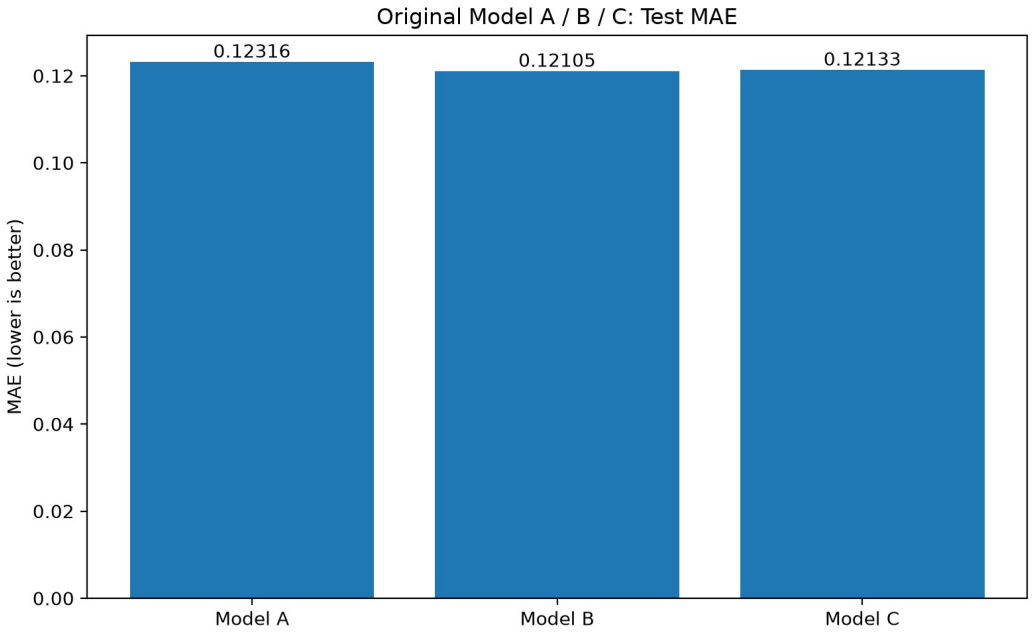


Figure 1: Test MAE for the principal Model A/B/C comparison. Lower values indicate lower average absolute error.

```
[1] ORIGINAL MODEL A / B / C
  Model  Test_N      MAE     RMSE       R2  MAE_improvement_vs_A_pct
Model A    1194 0.123159 0.172344 0.551043                  0.000000
Model B    1194 0.121047 0.169027 0.568156                  1.714856
Model C    1194 0.121325 0.169764 0.564382                  1.489132

[2] CORRECTED AUDIT (SEPARATE TEST SET)
                 Variant  Test_N      MAE     RMSE       R2  MAE_improvement_vs_core_pct
                    Core    1223 0.120966 0.170774 0.554984                     0.000000
                  Mode A    1223 0.119102 0.167612 0.571309                     1.540929
           Full enhanced    1223 0.118346 0.166695 0.575988                     2.165898
 Without tactical/manager    1223 0.118345 0.166290 0.578044                     2.166725
Without squad competition    1223 0.118494 0.166298 0.578005                     2.043549
            Without coach    1223 0.119102 0.167612 0.571309                     1.540929
```

Figure 2: Terminal output showing the principal A/B/C metrics and a separate corrected-audit experiment.

5.2 Paired statistical comparisons

The supplied paired-comparison tables report the following results. The paired mean difference is defined as the first model's absolute error minus the second model's absolute error, so positive values favour the second model.

Table 3: Paired MAE comparisons from supplied statistical outputs

Comparison	**Mean difference**	**Bootstrap 95% CI**	**Wilcoxon p**
A vs B	0.002112	[0.000764, 0.003406]	0.001579
A vs C	0.001834	[0.000192, 0.003431]	0.019487
B vs C	−0.000278	[−0.001564, 0.001006]	0.422961

For A versus B, the confidence interval excludes zero and the Wilcoxon p-value is below 0.05. Given the stated difference definition, this supports lower paired absolute error for Model B on the common test sample. For A versus C, the reported interval also excludes zero and the p-value is below 0.05, indicating evidence of lower error for Model C than A under the supplied analysis. For B versus C, the interval includes zero and the p-value is 0.422961. The analysis therefore does not show a statistically significant difference in paired absolute errors between these two models.

Failure to reject a difference is not proof that the models are equivalent. In addition, statistical significance does not by itself establish that the magnitude of an improvement is operationally important.

5.3 Feature importance

The supplied Random Forest feature-importance figure ranks pre-transfer goals per 90, position, pre-transfer goals, pre-transfer assists per 90, and other prior-performance and player attributes among the leading features. This is consistent with the model using recent player performance and player profile information strongly in its predictions.

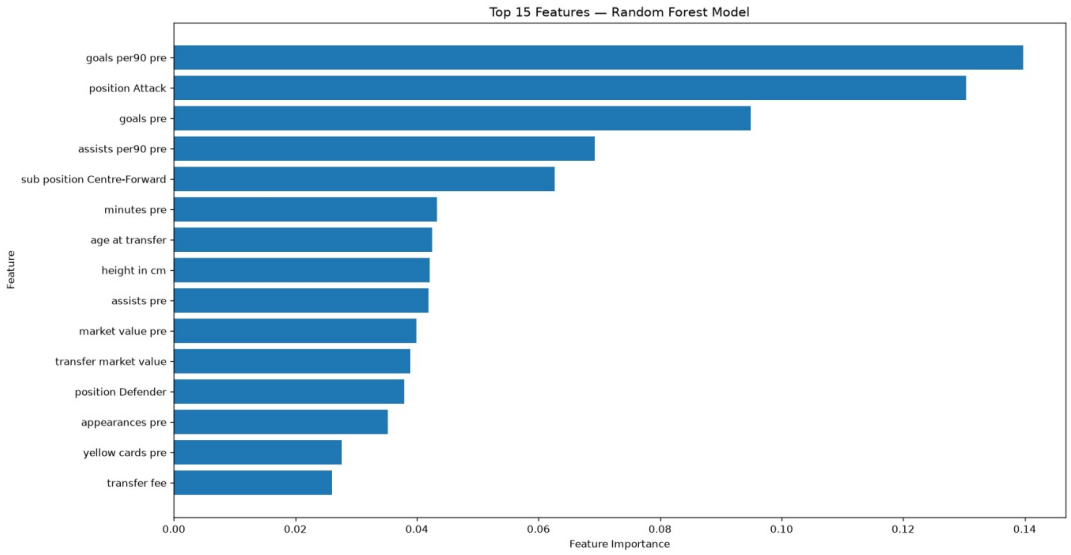


Figure 3: Top 15 features in the supplied Random Forest feature-importance figure.

Feature importance is descriptive of how a fitted model uses information. It does not establish that a feature causes post-transfer performance, and correlated predictors can share or redistribute importance.

5.4 Separate corrected-audit experiment

A separate corrected-audit screenshot reports 1,223 test records and compares six feature variants. These figures are included for completeness but are not merged with the principal A/B/C results because the test cohort and experimental setup differ.

Table 4: Reported corrected-audit results (separate experiment)

Variant	**Test** N	**MAE**	**RMSE**	R^2
Core	1,223	0.120966	0.170774	0.554984
Mode A	1,223	0.119102	0.167612	0.571309
Full enhanced	1,223	0.118346	0.166695	0.575988
Without tactical/manager	1,223	0.118345	0.166290	0.578044
Without squad competition	1,223	0.118494	0.166298	0.578005
Without coach	1,223	0.119102	0.167612	0.571309

These audit values are transcribed from the supplied terminal screenshot and have not been independently recalculated from the corresponding prediction file. They should therefore be labelled screenshot-reported results until the audit predictions are available for verification.

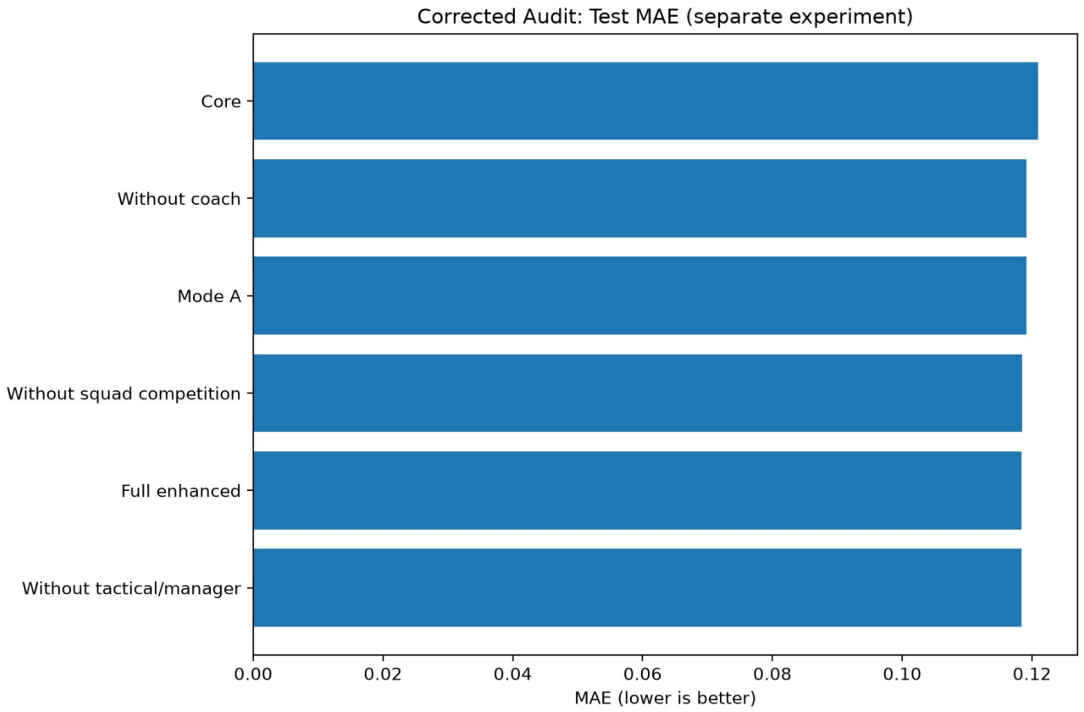


Figure 4: Corrected-audit test MAE from the separate experiment.

The full-enhanced and ablation variants have very similar reported errors. This may indicate that some feature groups contribute little additional predictive information in this particular audit, but the result should not be interpreted as proof that tactical or manager information is universally unhelpful. The audit requires its own uncertainty analysis and verification.

Chapter 6

Discussion

6.1 Basic destination-team context adds predictive value

The principal A/B/C comparison favours Model B. Its MAE is lower than Model A's by approximately 1.71%, while RMSE also decreases and R^2 increases. The paired statistical outputs provide evidence that this is not only a difference between aggregate metrics: the reported A–B confidence interval excludes zero and the Wilcoxon p-value is 0.001579.

The improvement is modest in absolute terms, so the conclusion should be appropriately qualified. The results suggest that basic destination-team context contains information useful for predicting post-transfer attacking output beyond the player-only feature set. They do not demonstrate that team context explains all or most of the uncertainty in individual outcomes.

6.2 Richer context does not improve on the basic model in this experiment

Model C performs better than Model A but slightly worse than Model B on MAE and RMSE. The B–C paired comparison is not statistically significant. This indicates that the richer feature configuration did not demonstrate an additional predictive gain in this experiment. Possible explanations include redundant features, noisy contextual variables, limited coverage, or insufficiently precise representations of the player's actual role. These are possible explanations, not conclusions established by the current results.

6.3 Interpretation of feature importance

The feature-importance figure places pre-transfer performance and player attributes among the most important predictors. This is plausible for a target based on subsequent goals and assists per 90 minutes, but importance rankings should be interpreted cautiously. They depend on the fitted model, feature correlation, data coverage, and the importance method. They are not causal explanations.

6.4 Practical implications

For research and exploratory scouting support, the findings suggest that a compact set of destination-team performance indicators may improve predictions relative to a player-only baseline. The evidence does not justify using the model as an autonomous transfer decision system. A practical deployment would require additional validation, calibration of uncertainty, checks across leagues and seasons, and assessment of how predictions change for players with limited prior minutes or unusual transfer contexts.

6.5 Relationship to prior work

The findings are consistent with the broader motivation in football analytics to combine individual performance information with team and tactical context [5, 7]. However, this project evaluates a different outcome from transfer-fee estimation [4] and player-action valuation [2]. Its specific contribution is the empirical comparison of feature configurations for post-transfer attacking performance, not a claim of priority over all prior work in football prediction.

Chapter 7

Limitations and Future Work

7.1 Limitations

1. **Dataset verification.** The final consolidated dataset was not included in the available support archive. The reported overall dataset size and target summary therefore remain to be independently checked.

2. **Feature timing and leakage.** The timing of each contextual feature should be audited to confirm that it was available at the intended prediction time. Any feature constructed using post-transfer information could invalidate the evaluation.

3. **Temporal split reproducibility.** The main script specifies a chronological 60/20/20 split, but the final dataset and exact split boundaries have not been independently reconstructed in this report.

4. **Hyperparameter selection.** The supplied description of the principal script does not establish that validation-based hyperparameter selection was used for the reported main A/B/C metrics. This must be confirmed from the exact script version that generated the saved predictions.

5. **Statistical procedure.** The supplied paired comparison outputs are consistent with the reported results, but the bootstrap resampling unit, confidence-interval procedure, and any multiple-comparison adjustment need confirmation. Repeated transfers for the same player may affect independence assumptions.

6. **Generalisation gap.** A separate Model A tuning experiment reported much higher training than validation/test fit. This indicates a generalisation gap in that separate experiment, but its precise cause is not established and its metrics must not be merged with the main comparison.

7. **Coverage and measurement.** Team-context coverage is limited by cross-source matching and data availability. Goals and assists per 90 minutes also provide only a narrow representation of player performance.

8. **Unobserved influences.** Injuries, playing time, role changes, teammate quality, adaptation, and manager decisions may affect post-transfer outcomes but may not be fully represented in the available features.

9. **Audit experiment.** The corrected-audit results are currently screenshot-reported and have not been independently recalculated from saved predictions.

7.2 Future work

Future work should first complete the reproducibility checks rather than add new model modules. Priority steps are to verify the final dataset summary, archive the exact script version used for the main comparison, audit feature availability dates, and reproduce the paired statistical analysis. Further evaluation could then examine performance across seasons, leagues, positions, and levels of prior playing time. If repeated-player dependence is present, uncertainty estimation could use an appropriate grouped resampling procedure. Additional context features should be retained only when they are available at prediction time and provide evidence of out-of-sample benefit.

Chapter 8

Conclusion

This project investigated whether destination-team context improves prediction of football players' post-transfer attacking performance, measured by `post_GA90`. In the principal comparison of 1,194 aligned test records, Model B, which combines player characteristics with basic destination-team context, achieved the lowest MAE (0.121047) and RMSE (0.169027), and the highest R^2 (0.568156). Compared with the player-only Model A, Model B reduced MAE by approximately 1.71%. The supplied paired statistical analysis supports a lower absolute error for Model B relative to Model A.

The richer Model C also outperformed Model A on the reported aggregate metrics, but it did not improve on Model B. The paired B–C comparison was not statistically significant ($p = 0.422961$), so the experiment does not establish an additional predictive benefit from the richer contextual feature set.

The central conclusion is therefore specific: basic destination-team context improved prediction in the reported main experiment, whereas richer context did not demonstrate a further statistically detectable improvement over the basic-context model. This is evidence from one experimental design, not proof that richer context is universally unhelpful. Final claims remain subject to verification of the dataset, feature timing, training procedure, and statistical resampling method. The model should be treated as an analytical research tool rather than a validated autonomous system for transfer decisions.

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Appendix A

Additional Output Screenshots

This appendix collects the supplied project output figures as supporting implementation and evaluation evidence. The principal A/B/C results are discussed in Chapter 5. The corrected-audit figures refer to a separate experiment and are not combined with the principal test cohort.

A.1 Principal model-comparison output

```
[1] ORIGINAL MODEL A / B / C
  Model  Test_N      MAE     RMSE       R2  MAE_improvement_vs_A_pct
Model A    1194 0.123159 0.172344 0.551043                  0.000000
Model B    1194 0.121047 0.169027 0.568156                  1.714856
Model C    1194 0.121325 0.169764 0.564382                  1.489132

[2] CORRECTED AUDIT (SEPARATE TEST SET)
                 Variant  Test_N      MAE     RMSE       R2  MAE_improvement_vs_core_pct
                    Core    1223 0.120966 0.170774 0.554984                     0.000000
                  Mode A    1223 0.119102 0.167612 0.571309                     1.540929
           Full enhanced    1223 0.118346 0.166695 0.575988                     2.165898
 Without tactical/manager    1223 0.118345 0.166290 0.578044                     2.166725
Without squad competition    1223 0.118494 0.166298 0.578005                     2.043549
           Without coach    1223 0.119102 0.167612 0.571309                     1.540929
```

Figure 5: Supplied terminal screenshot containing model-comparison output. The principal A/B/C experiment and the corrected audit are separate experiments.

A.2 Principal A/B/C test MAE

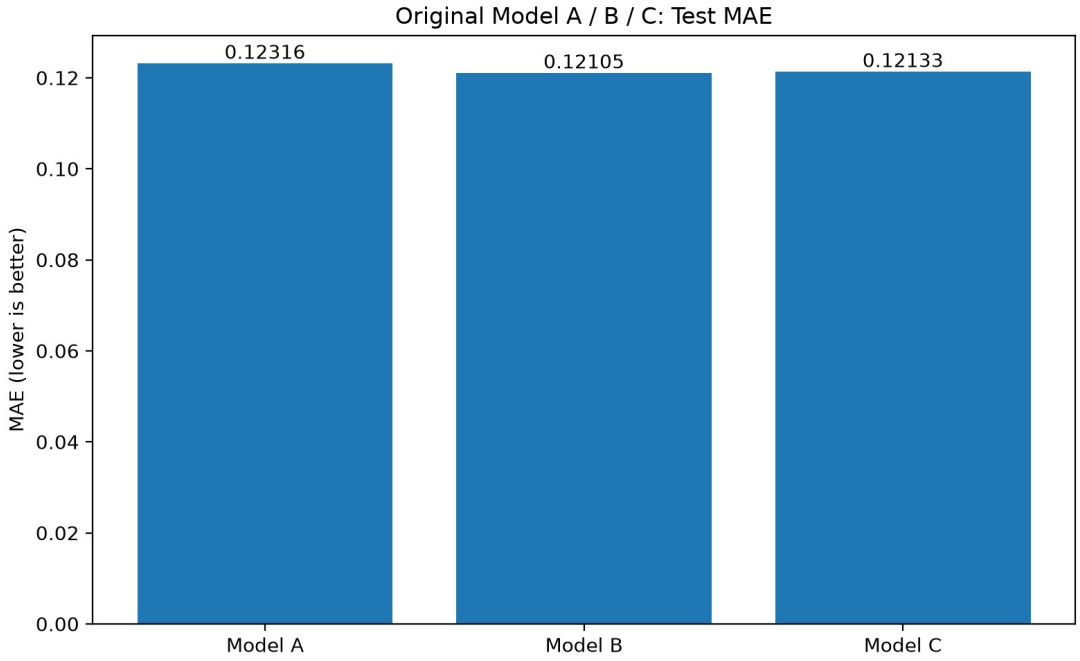


Figure 6: Supplied graph comparing test MAE for Models A, B, and C in the principal experiment.

A.3 Random Forest feature importance

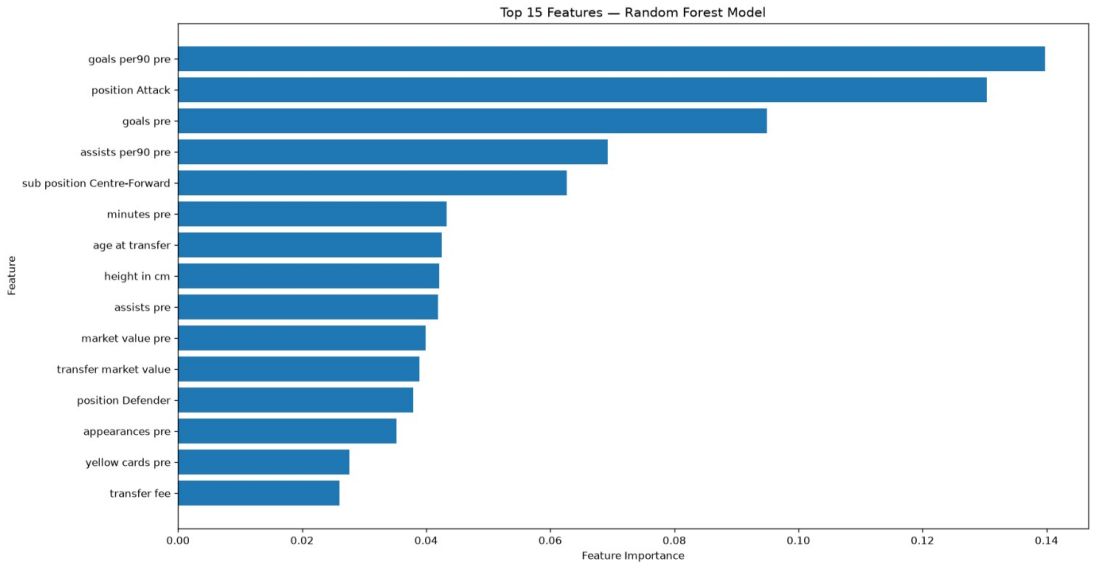


Figure 7: Supplied top-15 feature-importance figure. Feature importance is descriptive and does not establish causation.

A.4 Separate corrected-audit output

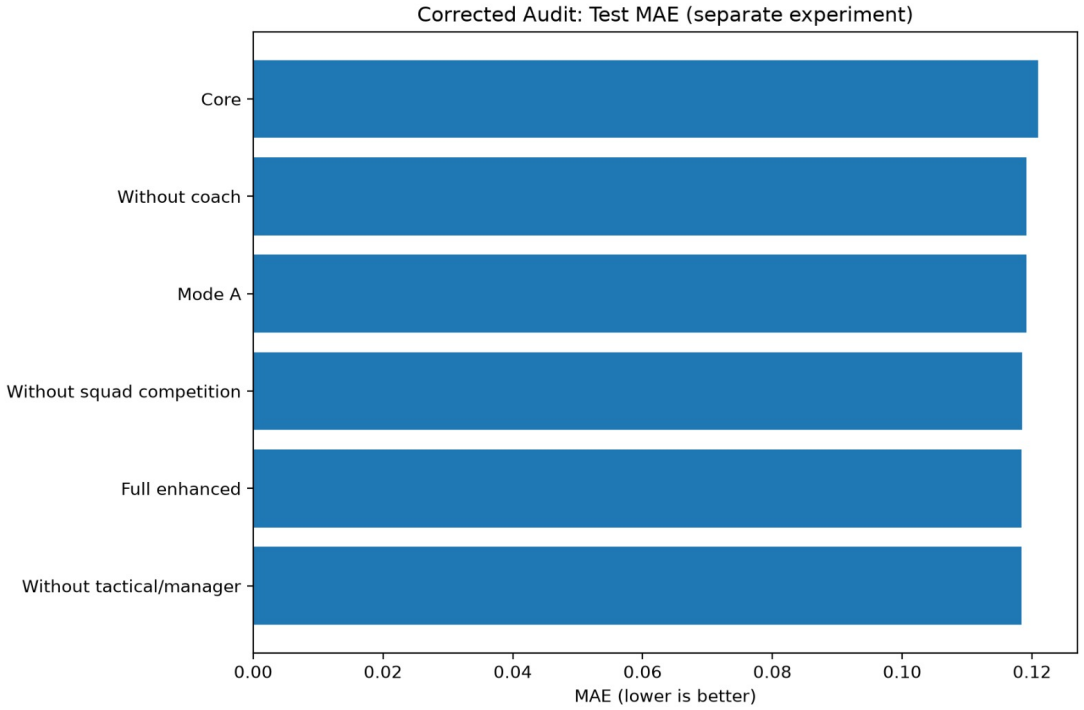


Figure 8: Supplied corrected-audit MAE graph. This experiment uses a different test cohort from the principal A/B/C comparison.

A.5 Prediction application: input workflow and output

The following selected screenshots document the prototype interface in workflow order. The form is organised into player profile, previous performance and transfer economics, destination-team context, tactical/formation context, squad competition and coach context. Only a small selection is included to demonstrate the workflow without overloading the appendix. The displayed example values and prediction are synthetic software-test inputs, not verified statistics for real players.

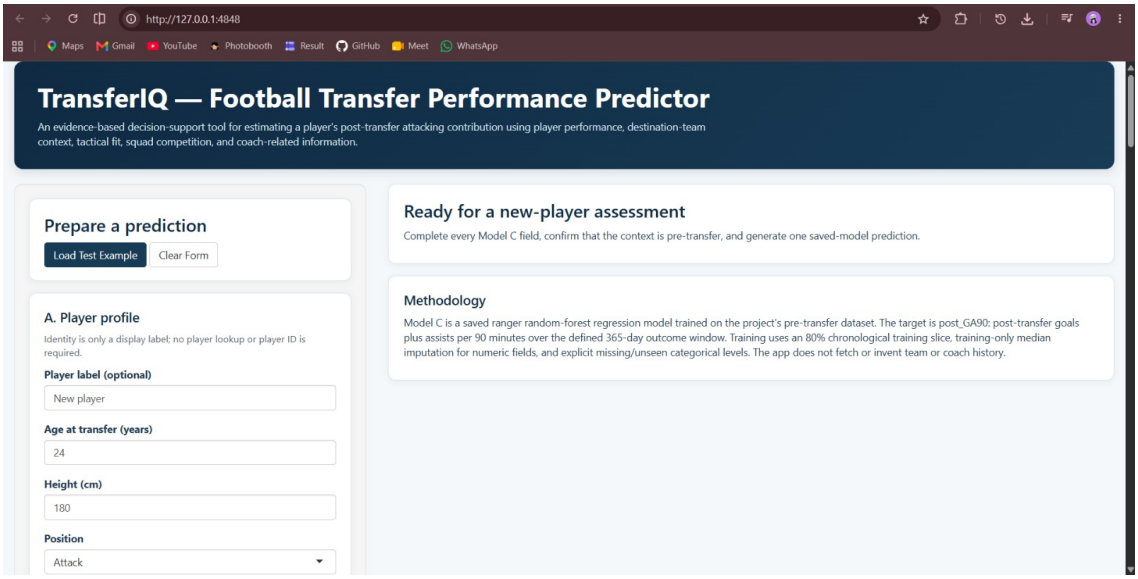
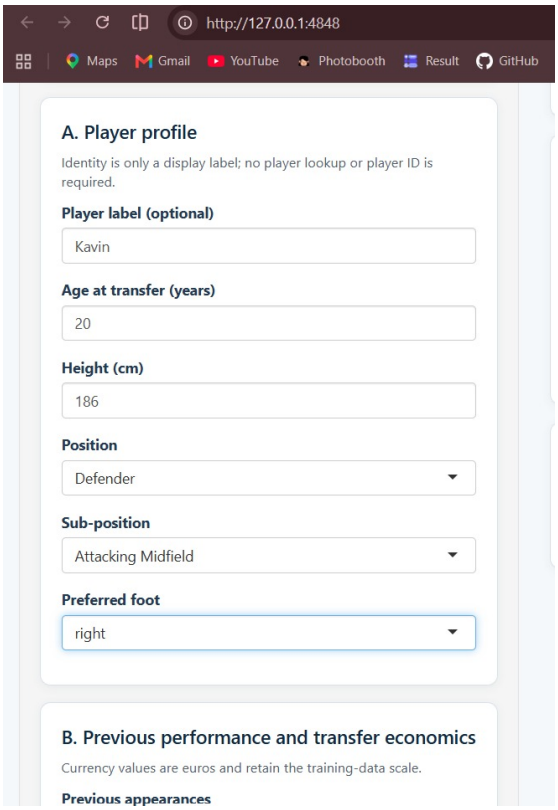
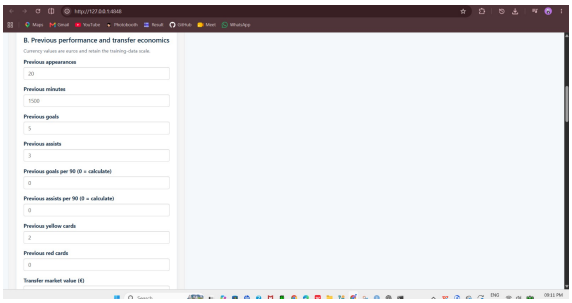


Figure 9: Application overview showing the prediction form and methodology panel before a new assessment.

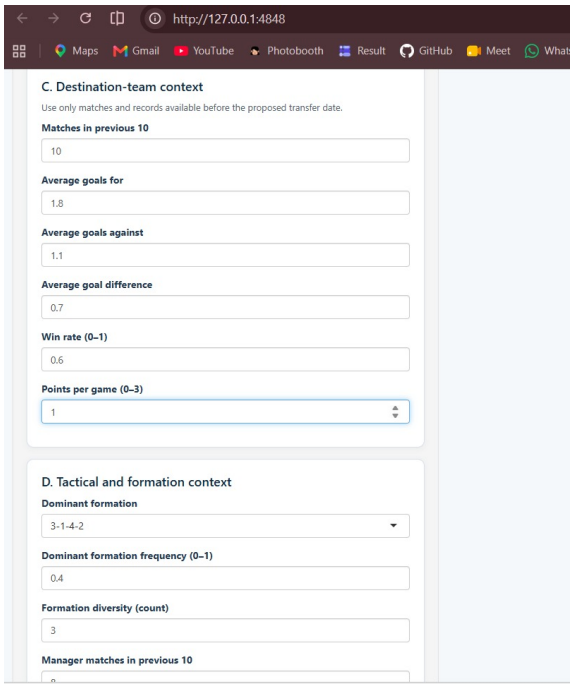


(a) Player profile fields.

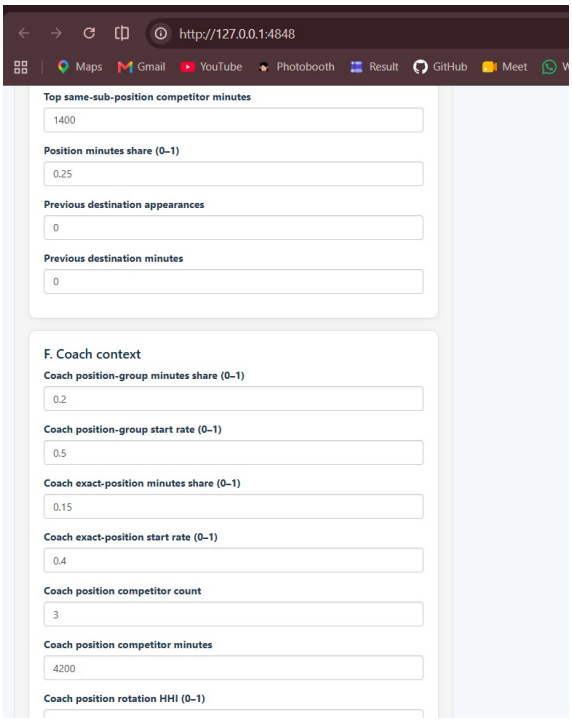


(b) Previous performance and transfer economics.

Figure 10: First stage of the input workflow.



(a) Destination-team and tactical context.



(b) Squad competition and coach context.

Figure 11: Additional contextual inputs required for the richer Model C prediction.

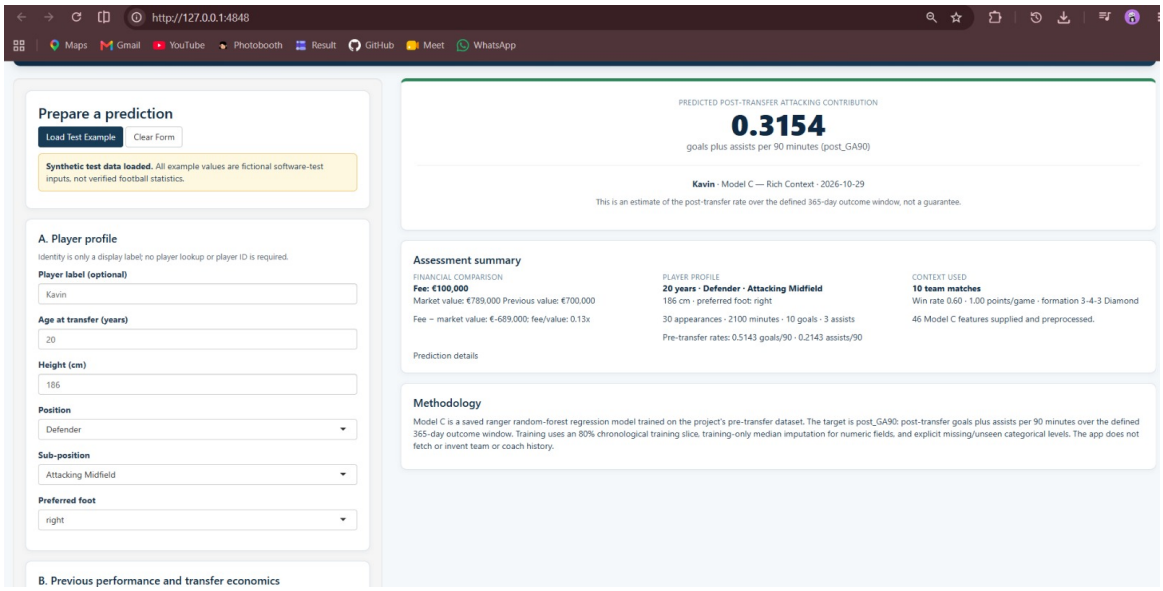


Figure 12: Example generated prediction and assessment summary. The displayed value is from synthetic test data and is included only as evidence of application behaviour, not as a real-player prediction.